LAPORAN PRAKTIKUM

POSTTEST 2

ALGORITMA PEMROGRAMAN LANJUT

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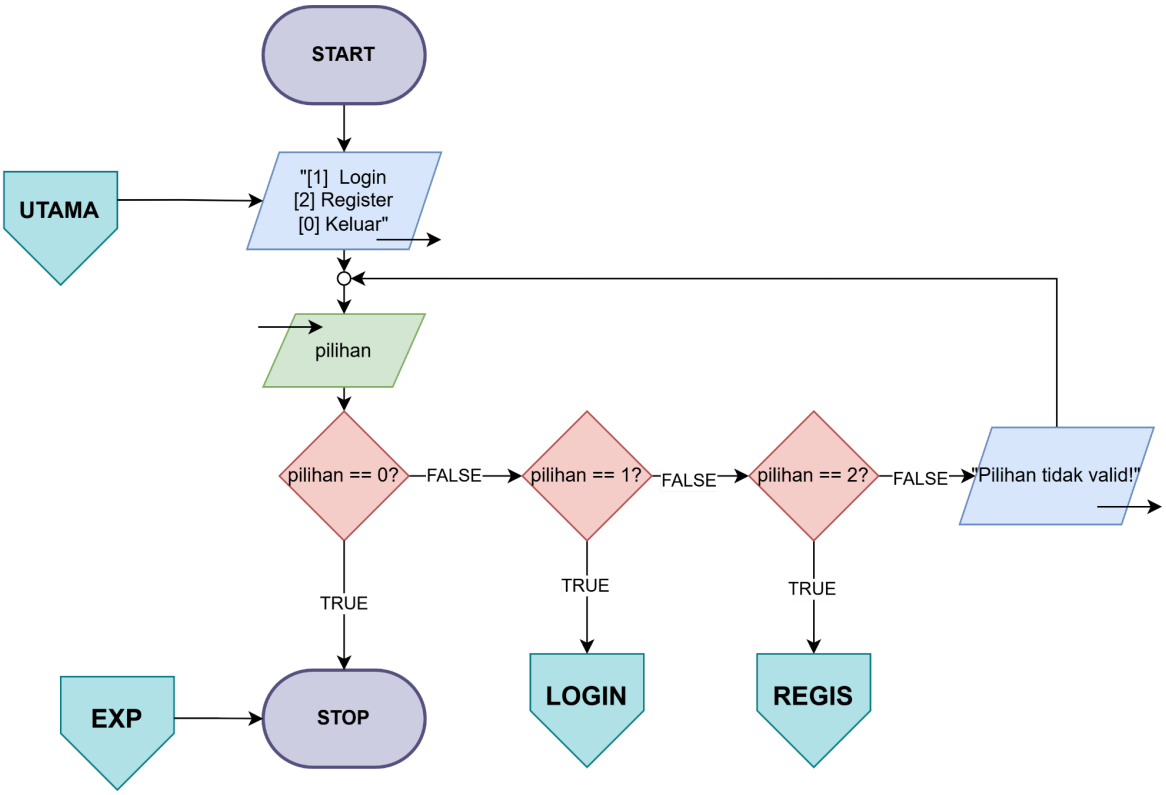
PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

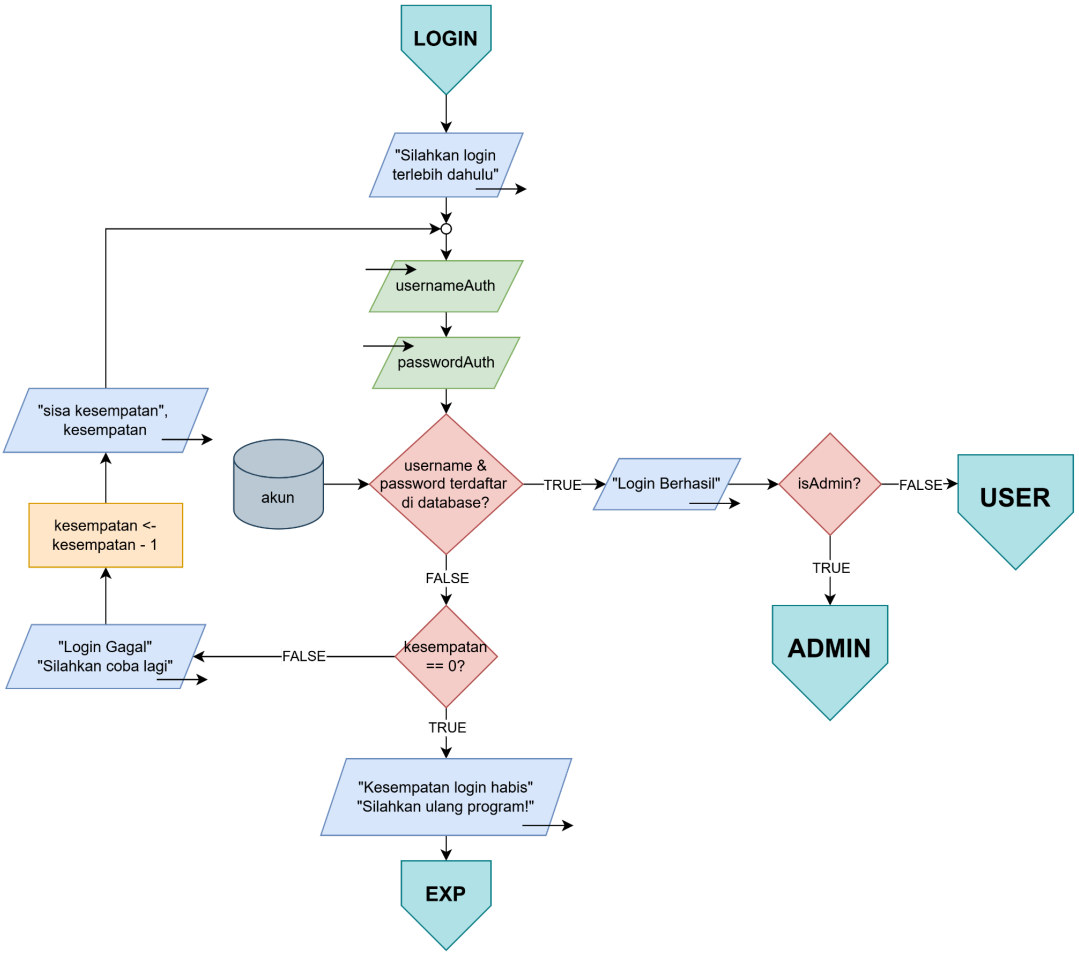
SAMARINDA

2026

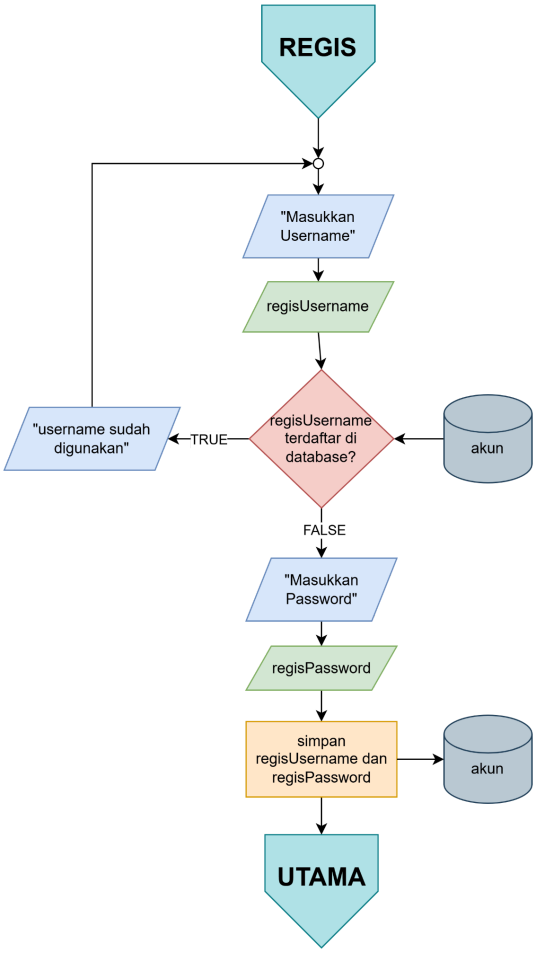
1. Flowchart



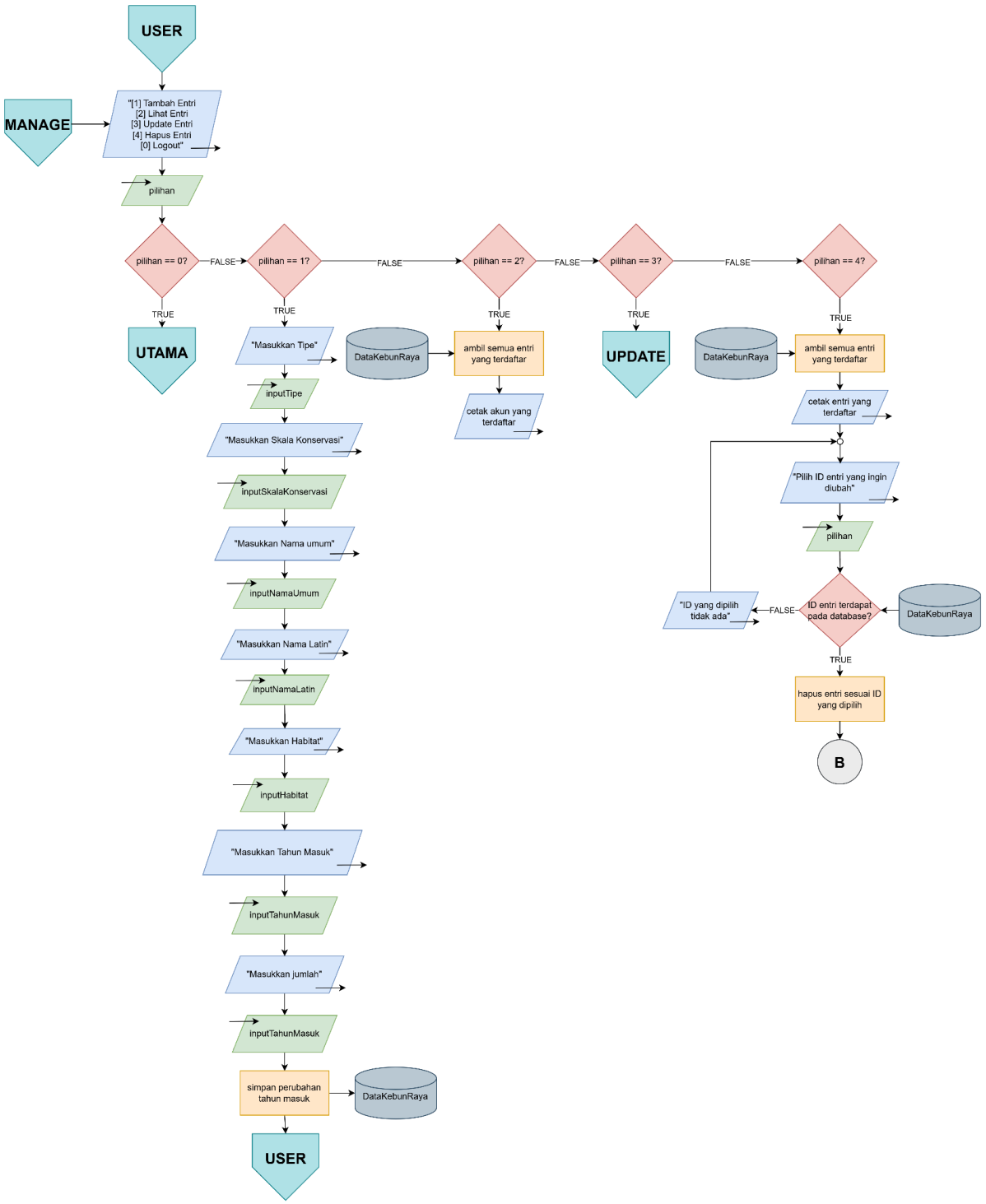
Gambar 1.1 – Menu Utama



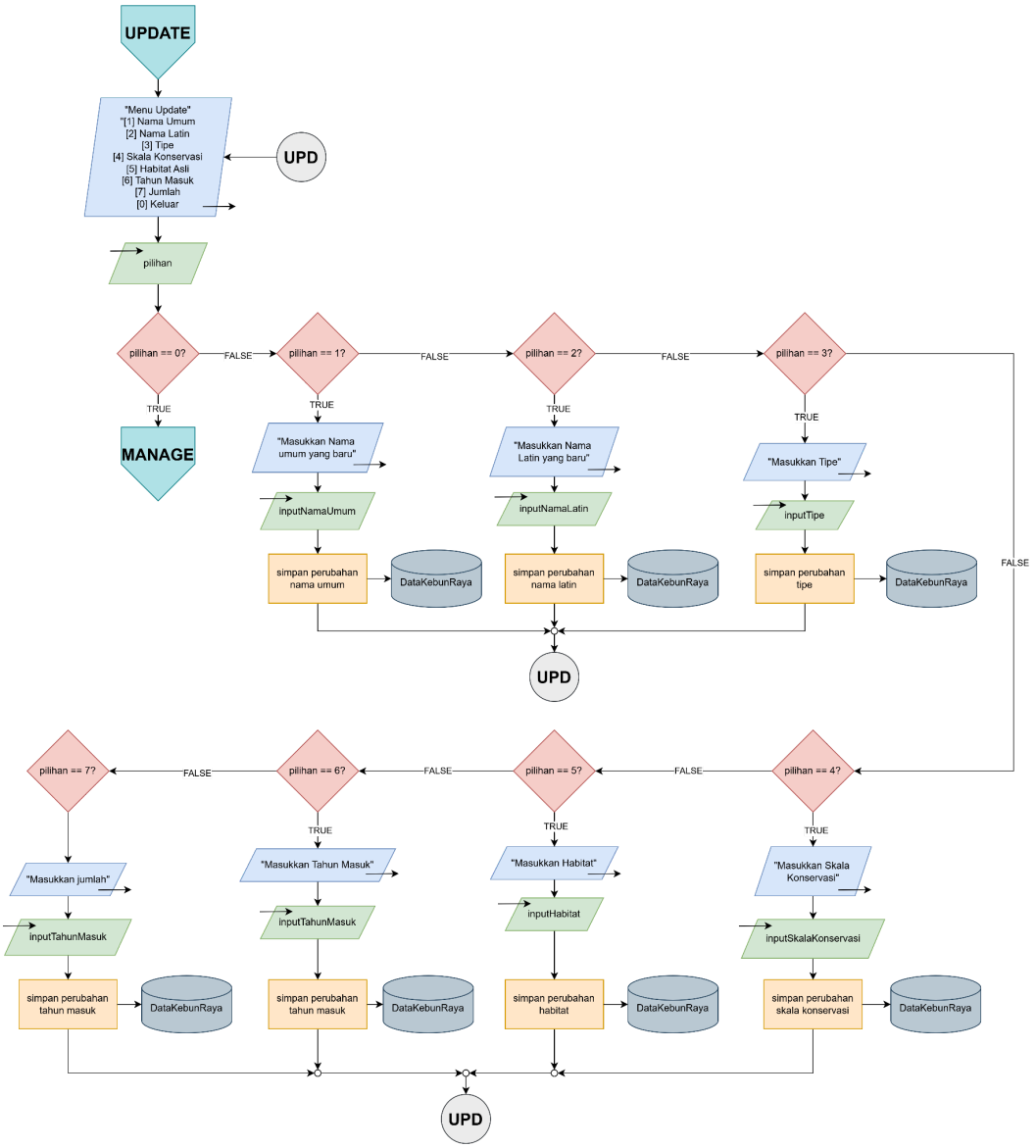
Gambar 1.2 – Menu Login



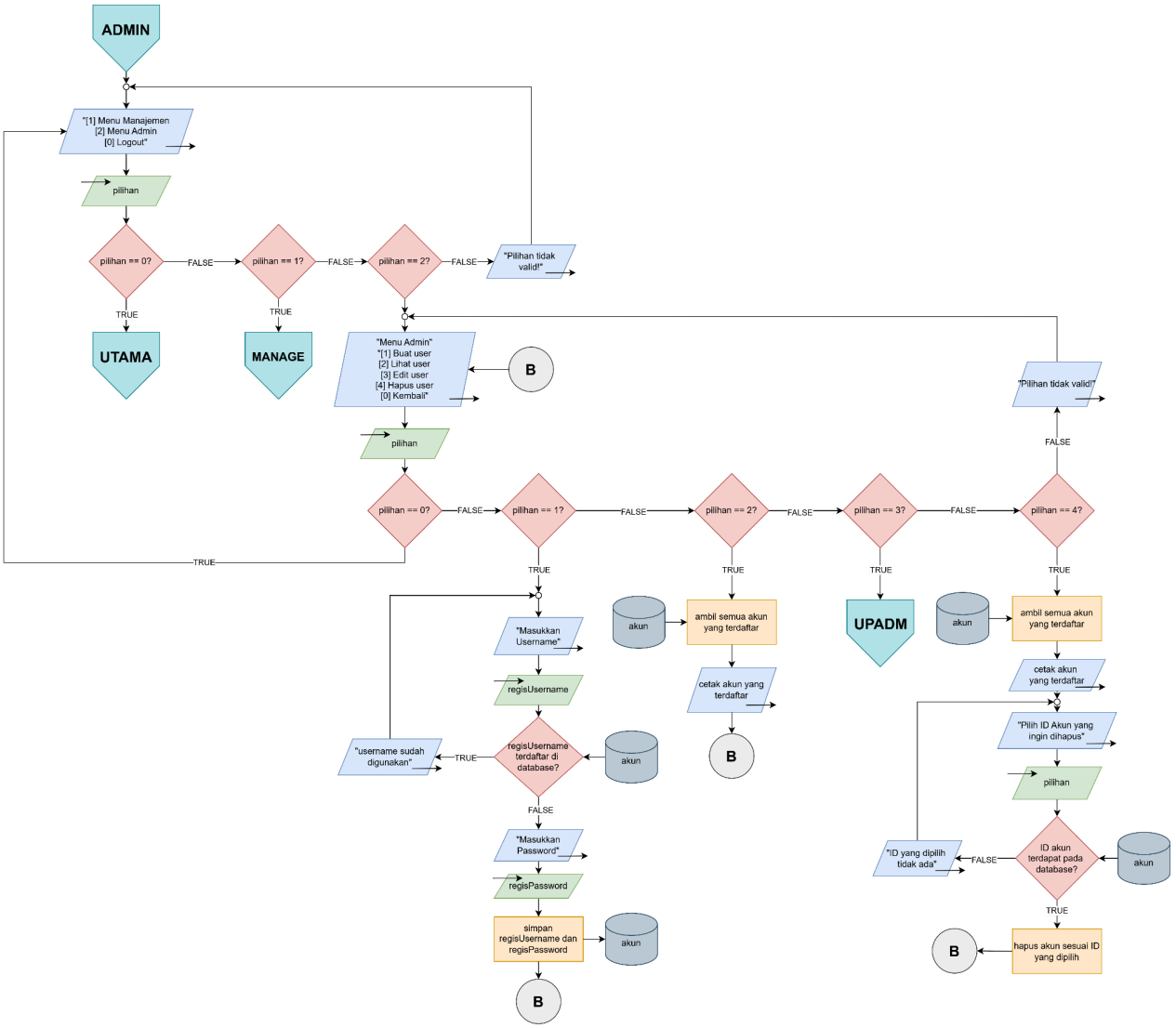
Gambar 1.3 – Menu Register



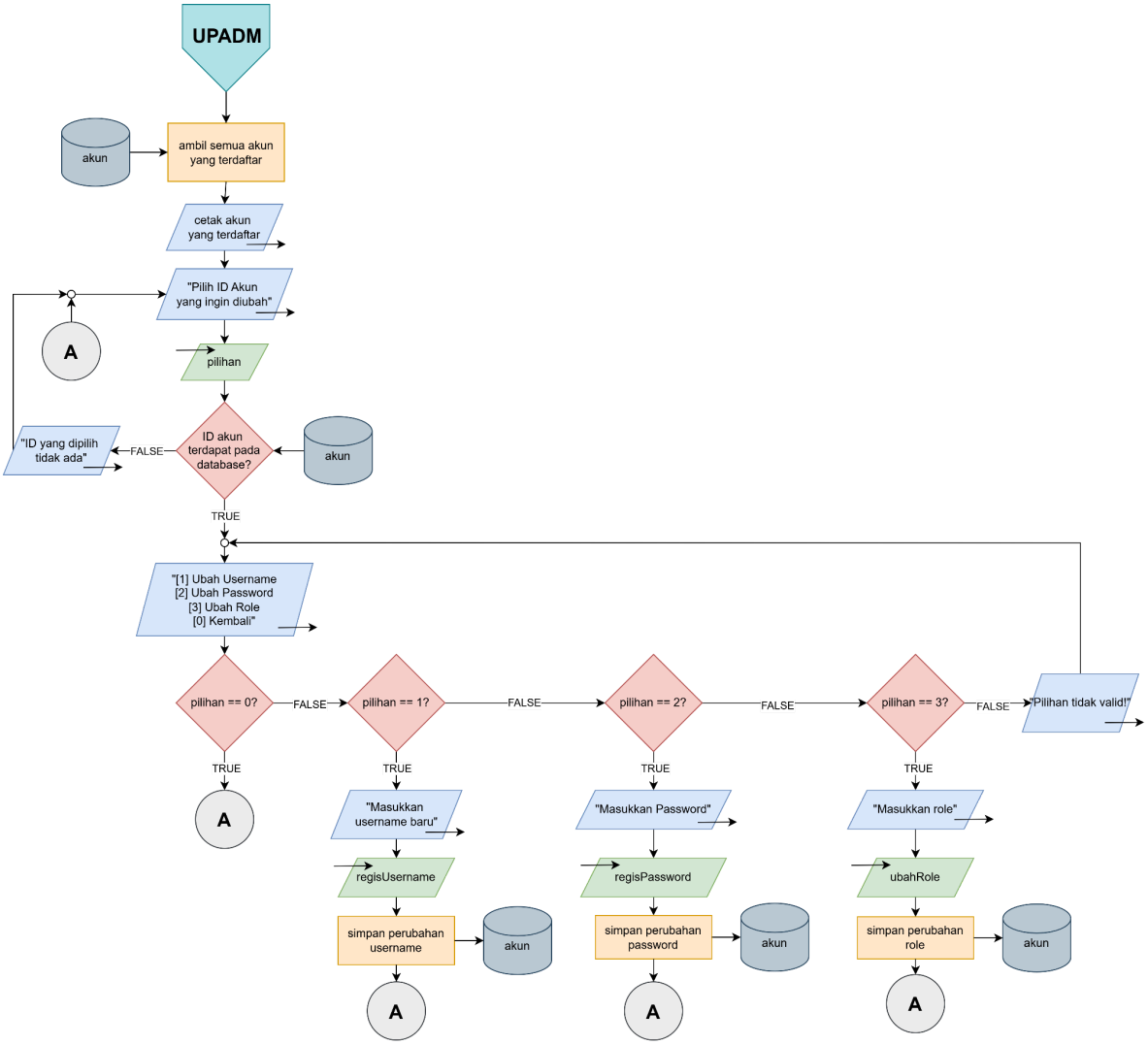
Gambar 1.4 – Menu Khusus User



Gambar 1.5 – Menu Update User



Gambar 1.6 – Menu Khusus Admin



Gambar 1.7 – Menu Update Admin

2. Deskripsi Singkat Program

Program ini terdiri dari tiga menu, yaitu:

2.1 Menu Utama

Menu Utama merupakan tampilan awal yang akan muncul saat program dijalankan. Pada menu ini, pengguna diberikan dua pilihan utama, yaitu **Login** dan **Register**. Fitur **Login** digunakan oleh pengguna yang sudah memiliki akun agar dapat masuk ke dalam sistem sesuai perannya, baik sebagai user maupun admin. Sementara itu, fitur **Register** digunakan untuk membuat akun baru dengan memasukkan data yang dibutuhkan

2.2 Menu Khusus User

Menu Khusus User adalah bagian program yang ditujukan untuk pengguna biasa setelah berhasil login. Pada menu ini, user dapat mengakses berbagai fitur yang berkaitan dengan pengelolaan data atau informasi kebun raya sesuai kebutuhan yang telah disediakan oleh program. Pengelolaan data dan informasi tersebut meliputi **nama umum**, **nama latin**, **skala konservasi**, **habitat**, **tipe**, dan lain **sebagainya**.

2.3 Menu Khusus Admin

Menu Khusus Admin merupakan menu yang hanya dapat diakses oleh pengguna dengan hak akses administrator. Pada bagian ini, admin memiliki kewenangan lebih besar dibandingkan user biasa, seperti **melihat akun-akun yang terdapat di database**, **membuat akun baru**, **mengedit akun-akun yang sudah ada**, dan **menghapus akun-akun dari database**.

3. Source Code

3.1. Menu Login

Source Code:

```
// --- Ada Kode di bagian atas ---

else if (pilihan == 1) { // MENU LOGIN
    cin.ignore();
    while (true) {
        cout << "\n" << GARIS_BESAR << endl << "MENU LOGIN" << endl <<
GARIS_BESAR << endl;
        cout << "Username : "; getline(cin, usernameAuth);
        cout << "Password : "; getline(cin, passwordAuth);

        for (int i = 0; i < totalPengguna; i++) { // DETEKSI AKUN
            if (akun[i].username == usernameAuth & akun[i].password ==
passwordAuth) {
                berhasilLogin = true;
                indeksUserSekarang = i;
                break;
            }
        }

        if (berhasilLogin) {
            cout << GARIS_SEDANG << endl << "Login berhasil. Selamat datang!" <<
endl;
            break;
        } else {
            kesempatanLogin--;
            if (kesempatanLogin > 0) {
                cout << GARIS_SEDANG << endl << "Login gagal. Silakan coba
lagi." << endl;
                cout << "Sisa kesempatan : " << kesempatanLogin << endl;
            } else {
                cout << GARIS_SEDANG << endl << "Kesempatan login habis!" <<
endl;
                cout << "Silakan mulai ulang program!" << endl;
                cout << endl;
                exit(1);
                break;
            }
        }
    }
}

// --- Ada Kode di bagian bawah ---
```

3.2. Menu Register

Source Code:

```
// --- Ada Kode di bagian atas ---

else if (pilihan == 2) { // MENU REGISTER
    cin.ignore();
    while(true) {
        bool usernameValid = false;

        while (!usernameValid) {
            cout << "\n" << GARIS_BESAR << endl << "MENU REGISTER" << endl <<
GARIS_BESAR << endl;
            cout << "Username baru : ";
            getline(cin, regisUsername);

            usernameValid = true;
            for (int i = 0; i < totalPengguna; i++) {
                if (regisUsername == akun[i].username) {
                    cout << "Username sudah dipakai. Silakan gunakan username
lain." << endl;
                    usernameValid = false;
                }
            }
        }

        cout << "Password baru : ";
        getline(cin, regisPassword);

        cout << GARIS_SEDANG << endl << "Akun berhasil dibuat untuk username: "
<< regisUsername << endl;

        akun[totalPengguna].userID = totalPengguna + 1;
        akun[totalPengguna].username = regisUsername;
        akun[totalPengguna].password = regisPassword;
        akun[totalPengguna].isAdmin = false;
        totalPengguna++;
        break;
    }
}

// --- Ada Kode di bagian bawah ---
```

3.3. Menu Khusus User

Source Code:

```
// --- Ada Kode di bagian atas ---

while (true) {
    cout << "\n" << GARIS_BESAR << endl << "MENU MANAJEMEN KEBUN RAYA" << endl
<< GARIS_BESAR << endl;
    if (adminToUser) {
        cout << "Selamat Datang Kembali Admin " <<
akun[indeksUserSekarang].username << endl;
    } else {
        cout << "Selamat Datang Kembali " << akun[indeksUserSekarang].username
<< endl;
    }
    cout << GARIS_BESAR << endl;
    cout << "[1] Tambah Entri Baru" << endl;
    cout << "[2] Lihat Entri" << endl;
    cout << "[3] Update Entri" << endl;
    cout << "[4] Hapus Entri" << endl;
    cout << GARIS_BESAR << endl;
    if (adminToUser) {
        cout << "[0] Kembali" << endl;
    } else {
        cout << "[0] Logout" << endl;
    }
    cout << GARIS_BESAR << endl;
    cout << "Pilih menu : "; cin >> pilihan;
}

// --- Ada Kode di bagian bawah ---
```

3.4. Menu Khusus Admin

Source Code:

```
// --- Ada Kode di bagian atas ---

while (true) {
    cout << "\n" << GARIS_BESAR << endl << "MENU AKSES ADMIN" << endl <<
GARIS_BESAR << endl;
    cout << "[1] Menu Manajemen" << endl;
    cout << "[2] Menu Admin" << endl;
    cout << GARIS_SEDANG << endl;
    cout << "[0] Logout" << endl;
    cout << GARIS_SEDANG << endl;
    cout << "Pilih menu : "; cin >> pilihan;

    if (pilihan == 0) {
        berhasilLogin = false;
        break;
    }
```

```
}

// --- Ada Kode di bagian bawah ---
```

4. Hasil Output

```
==================================================================
                SISTEM MANAJEMEN KEBUN RAYA
==================================================================
[1] Login
[2] Register
------------------------------------------------------------------
[0] Keluar
==================================================================
Pilihan Anda : 1
```

Gambar 4.1 – Menu Utama

```
==================================================================
MENU LOGIN
==================================================================
Username : Yoga
Password : 017
------------------------------------------------------------------
Login berhasil. Selamat datang!
```

Gambar 4.2 – Menu Login

```
==================================================================
MENU REGISTER
==================================================================
Username baru : user1
Password baru : 0123
------------------------------------------------------------------
Akun berhasil dibuat untuk username: user1
==================================================================
```

Gambar 4.3 – Menu Register

```
==================================================================
MENU MANAJEMEN KEBUN RAYA
==================================================================
Selamat Datang Kembali user1
==================================================================
[1] Tambah Entri Baru
[2] Lihat Entri
[3] Update Entri
[4] Hapus Entri
[4] Hapus Entri
==================================================================
[0] Logout
==================================================================
Pilih menu : 1
```

Gambar 4.4 – Menu Khusus User

```
==============================================================
TAMBAH ENTRI BARU
==============================================================
[1] Flora
[2] Fauna
==============================================================
[0] Kembali
==============================================================
Masukkan Pilihan Anda
> 1
Pilihan terpilih : 1

--------------------------------------------------------------
PILIH SKALA KONSERVASI
--------------------------------------------------------------
[1] Risiko Rendah (LC)
[2] Hampir Terancam (NT)
[3] Rentan (VU)
[4] Terancam (EN)
[5] Sangat Terancam (CR)
[6] Punah di Alam Liar (EW)
==============================================================
[0] Kembali
==============================================================
Masukkan Pilihan Anda
> 1
Masukkan Nama Umum: Amaryllis
Masukkan Nama Latin: Amaryllis belladona
Masukkan Habitat Asli: Yunani
Masukkan Tahun Masuk Kebun Raya: 2009
Masukkan Jumlah Specimen: 298
```

Gambar 4.5 – Create User

```
==============================================================
DAFTAR ENTRI KEBUN RAYA
==============================================================
+----+-----------+---------------------+-------+--------------------+-----------+-------+--------+
| ID | Nama Umum | Nama Latin          | Jenis | Skala              | Habitat   | Tahun | Jumlah |
+----+-----------+---------------------+-------+--------------------+-----------+-------+--------+
| 1  | Padi      | Oryza sativa        | Flora | Aman               | Indonesia | 2000  | 100000 |
+----+-----------+---------------------+-------+--------------------+-----------+-------+--------+
| 2  | Amaryllis | Amaryllis belladona | Flora | LC - Resiko Rendah | Yunani    | 2009  | 298    |
+----+-----------+---------------------+-------+--------------------+-----------+-------+--------+
```

Gambar 4.6 – Read User

```
==============================================================
UPDATE ENTRI
==============================================================
+----+-----------+-------+
| ID | Nama Umum | Jenis |
+----+-----------+-------+
| 1  | Padi      | Flora |
+----+-----------+-------+
| 2  | Amaryllis | Flora |
+----+-----------+-------+
==============================================================
[0] Keluar
==============================================================
Masukkan Pilihan Anda
> 1
---- MENU UPDATE ---------------------------------------------
[1] Nama Umum
[2] Nama Latin
[3] Tipe
[4] Skala Konservasi
[5] Habitat Asli
[6] Tahun Masuk
[7] Jumlah
--------------------------------------------------
[0] Keluar
--------------------------------------------------
Masukkan Pilihan Anda
> 4
```

Gambar 4.7 – Update User

```
==============================================================
HAPUS ENTRI
==============================================================
+----+-----------+-------+
| ID | Nama Umum | Jenis |
+----+-----------+-------+
| 1  | Padi      | Flora |
+----+-----------+-------+
| 2  | Amaryllis | Flora |
+----+-----------+-------+
| 3  | m         | Fauna |
+----+-----------+-------+
==============================================================
[0] Keluar
==============================================================
Masukkan Pilihan Anda
> 3

--------------------------------------------------------------
ENTRI YANG AKAN DIHAPUS
Nama Umum: m
Nama Latin: m
Jenis: Fauna
Skala Konservasi: NT - Hampir Terancam
Habitat Asli: m
Tahun Masuk: m
Jumlah: 1009
-----------------------------------------------------------------
--- KONFIRMASI HAPUS ---
[1] Ya
[2] Tidak
----------------------------------------------
Masukkan Pilihan Anda
> 1
---- KONFIRMASI HAPUS ---
```

Gambar 4.8 – Delete User

```
==============================================================
MENU ADMIN
==============================================================
[1] Tambah User
[2] Lihat User
[3] Update User
[4] Hapus User
-------------------------------
[0] Keluar
-----------------------------
Masukkan Pilihan Anda
> 1
--- Menambahkan User Baru ---
Username baru : user2
Password baru : 987
```

Gambar 4.9 – Create Admin

```
==============================================================
DAFTAR USER
==============================================================
+----+----------+----------+-------+
| ID | Username | Password | Role  |
+----+----------+----------+-------+
| 1  | Yoga     | 017      | Admin |
+----+----------+----------+-------+
| 2  | user1    | 0123     | User  |
+----+----------+----------+-------+
| 3  | user2    | 987      | User  |
+----+----------+----------+-------+
```

Gambar 4.10 – Read Admin

```
==============================================================
UPDATE USER
==============================================================
+----+----------+--------+
| ID | Username | Role   |
+----+----------+--------+
| 1  | Yoga     | Admin  |
+----+----------+--------+
| 2  | user1    | Member |
+----+----------+--------+
| 3  | user2    | Member |
+----+----------+--------+
==============================================================
[0] Keluar
==============================================================
Masukkan Pilihan Anda
> 2
--- MENU UPDATE ---
[1] Ubah Username
[2] Ubah Password
[3] Ubah Role
---------------------------------
[0] Keluar
---------------------------------
Masukkan Pilihan Anda
> 3
--- MENGGANTI ROLE ---
[1] Admin
[2] User
-------------------------------
[0] Kembali
-------------------------------
Masukkan Pilihan Anda
> 1
```

Gambar 4.11 – Update Admin

```
==============================================================
HAPUS USER
==============================================================
+----+----------+--------+
| ID | Username | Role   |
+----+----------+--------+
| 1  | Yoga     | Admin  |
+----+----------+--------+
| 2  | user1    | Admin  |
+----+----------+--------+
| 3  | user2    | Member |
+----+----------+--------+
==============================================================
[0] Keluar
==============================================================
Masukkan Pilihan Anda
> 2

--------------------------------------------------------------
AKUN YANG AKAN DIHAPUS
ID Akun: 2
Username: user1
Role: Admin
-------------------------------------------------------
[1] Hapus
-------------------------------------------------------
[0] Kembali
-------------------------------------------------------
Masukkan Pilihan Anda
> 1
--------------------------------------------------------------
KONFIRMASI HAPUS
```

Gambar 4.12 – Delete Admin

5. Langkah-langkah GIT

5.3 GIT Add

```
PS E:\kuliah\Semester 2\APL\praktikum-apl> git add post-test
PS E:\kuliah\Semester 2\APL\praktikum-apl> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   post-test/post-test-apl-2/2509106017-YogaAnandaPrasetya-PT-2.cpp
        new file:   post-test/post-test-apl-2/2509106017-YogaAnandaPrasetya-PT-2.exe
```

Gambar 5.3 – Git add

"Git add" disini berfungsi untuk memasukkan file yang sudah kita kerjakan ke dalam staging area. Staging area disini berfungsi sebagai tempat singgah sementara file-file yang ingin kita commit di git nantinya. "Git status" disini hanya berfungsi untuk memeriksa apakah file yang ingin di-commit sudah ada di staging area atau belum.

5.4 GIT Commit

```
PS E:\kuliah\Semester 2\APL\praktikum-apl> git commit -m "feat: Upload Source Code"
[main 7f6fe66] feat: Upload Source Code
 2 files changed, 934 insertions(+)
 create mode 100644 post-test/post-test-apl-2/2509106017-YogaAnandaPrasetya-PT-2.cpp
 create mode 100644 post-test/post-test-apl-2/2509106017-YogaAnandaPrasetya-PT-2.exe
```

Gambar 5.4 – Git Commit

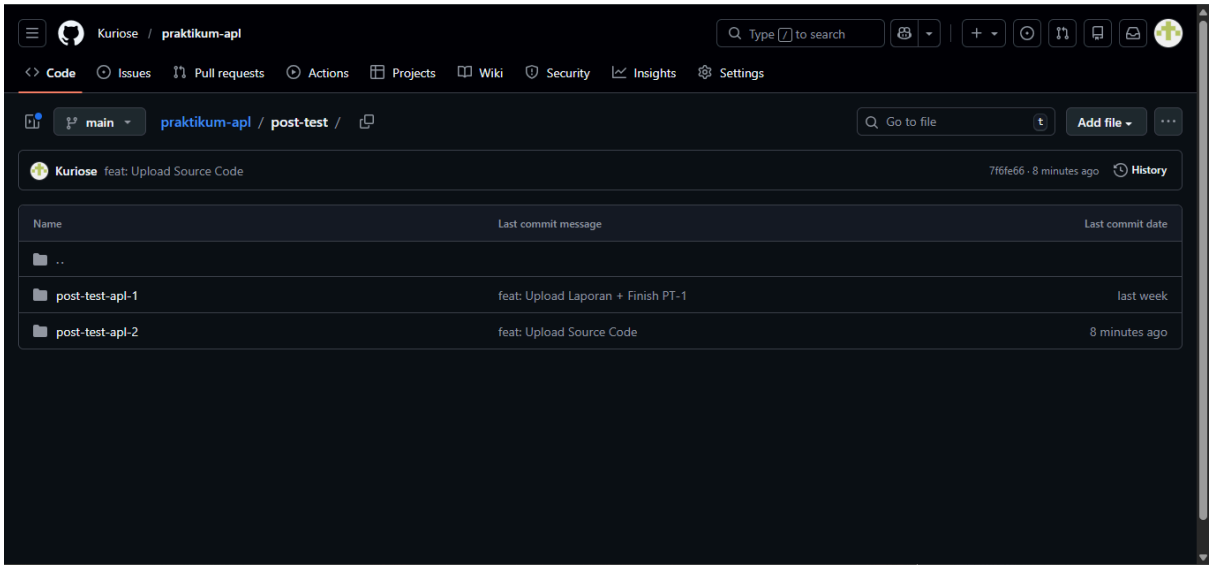
"Git Commit" disini berfungsi untuk menyimpan riwayat dari kode yang telah kita kerjakan ke dalam repository yang ada di dalam komputer kita (repository lokal). Disini kita juga dapat meninggalkan pesan singkat untuk mengingatkan kita atau orang lain terkait apa yang sudah kita kerjakan di kode tersebut.

5.5 GIT Push

```
PS E:\kuliah\Semester 2\APL\praktikum-apl> git push -u origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 824.81 KiB | 2.24 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Kuriose/praktikum-apl.git
   90ad7bd..7f6fe66  main -> main
branch 'main' set up to track 'origin/main'.
```

Gambar 5.5 – Git Push

"Git Push" disini berfungsi untuk mengunggah file-file yang sudah di-commit di repository lokal menuju ke repository yang ada di Github. Tujuannya adalah agar file-file yang sudah dikerjakan dapat diakses secara online dan dapat diakses oleh siapapun selama pengaturan repository-nya tidak di-private.



Gambar 5.6 – Tampilan pada web Github